

Resampling methods

The sampling distribution

- ▶ Ex: $X_1, X_2, \dots, X_n$ independent and identically distributed (iid) stochastic variables, drawn from $X \sim F$ where F is an unknown CDF
We would like to estimate $\mu = E(X)$ with $\bar{X}$
- ▶ Sampling distribution: Distribution of $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

The sampling distribution - An example

- ▶ X = height of a randomly chosen Norwegian adult
- ▶ Want to estimate the expected height, $\mu = E(X)$
- ▶ Draw a sample of $n = 100$, $x_1, x_2, \dots, x_n$
- ▶ Estimate $\hat{\mu} = \bar{x}$ (one single value)
- ▶ The sampling distribution of $\hat{\mu}$ is the distribution (histogram) of the $\bar{x}$ if we repeatedly sampled 100 observations.
- ▶ In this example we exemplify with estimating the parameter $\mu = E(X)$ but the arguments and concepts go through for any estimated quantity based on a random sample.

A sample of 100 observations

```
head(x,5)
```

```
## [1] 188.7829 190.6420 178.3605 173.1975 189.0883
```

```
mean(x)
```

```
## [1] 183.678
```

```
sd(x)
```

```
## [1] 16.72096
```

$\bar{x}_1 = 183.678$

Another sample of 100 observations

```
head(x,5)
```

```
## [1] 183.3589 162.6567 186.3363 160.1287 182.1163
```

```
mean(x)
```

```
## [1] 180.6785
```

```
sd(x)
```

```
## [1] 15.16861
```

$\bar{x}_2 = 180.678$

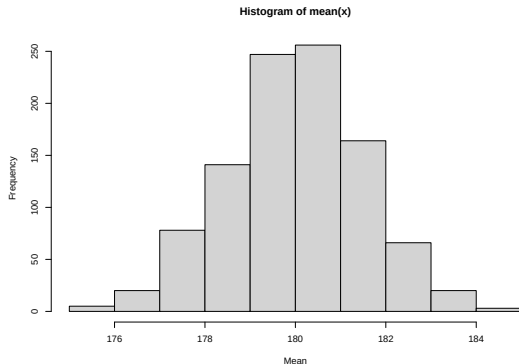
1000 different samples of 100 observations each

```
set.seed(12345)
xpop=rnorm(n=4000000,mean=180,sd=15)
n=100
nrepl=1000
res=matrix(0,nrepl,1)
for(irepl in 1:nrepl)
{
  x=sample(xpop,size = n)
  res[irepl]=mean(x)
}
hist(res,xlab="Mean",main="Histogram of mean(x)")
```

- ▶ Large sample approximation: The central limit theorem implies that the distribution of $\bar{X}$ can be approximated by $N\left(\mu, \frac{\sigma^2}{n}\right)$ for large samples
- ▶ $\therefore \bar{X}$ approx $N(180, 2.25)$
- ▶ In this case we know what μ and σ are since we have generated the population ourselves.

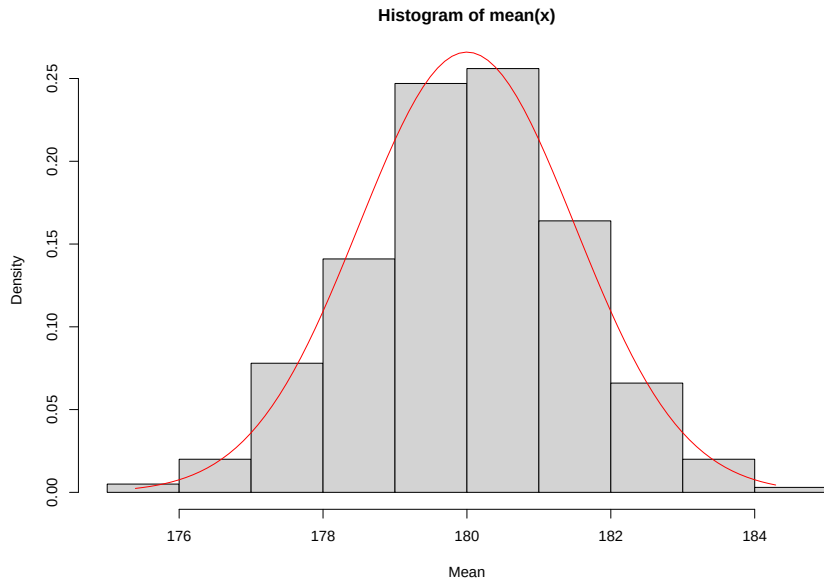
1000 different samples of 100 observations each

- ▶ Of course, this is not done in practice, we only do this as a mean to understand what a sampling distribution is.
- ▶ $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_{1000}$



With the large sample approximation

$$\therefore \bar{X} \sim N(180, 2.25)$$



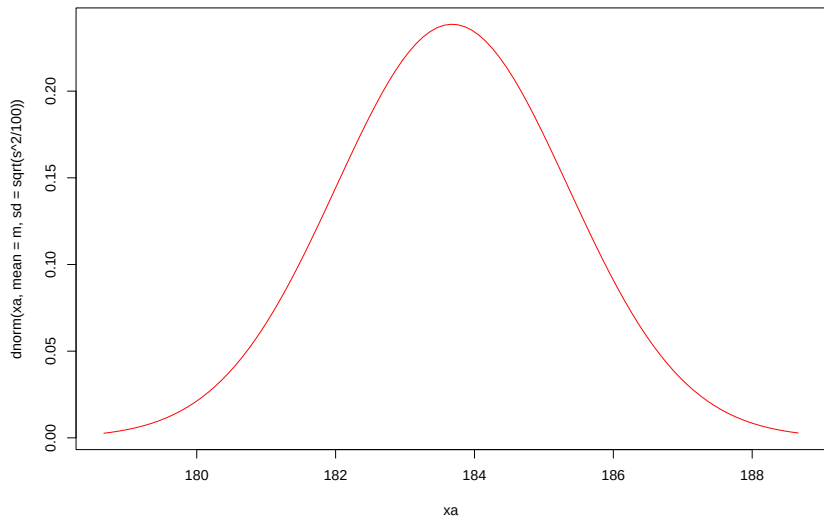
Can we estimate this distribution if we only have one single sample?

- ▶ Yes, there are several methods
- ▶ Large sample approximation: The central limit theorem implies that the distribution of $\bar{X}$ can be approximated by $N\left(\mu, \frac{\sigma^2}{n}\right)$ for large samples. Plug in estimators for μ and σ .
- ▶ Let us go back to the more realistic situation where only the first sample is available.

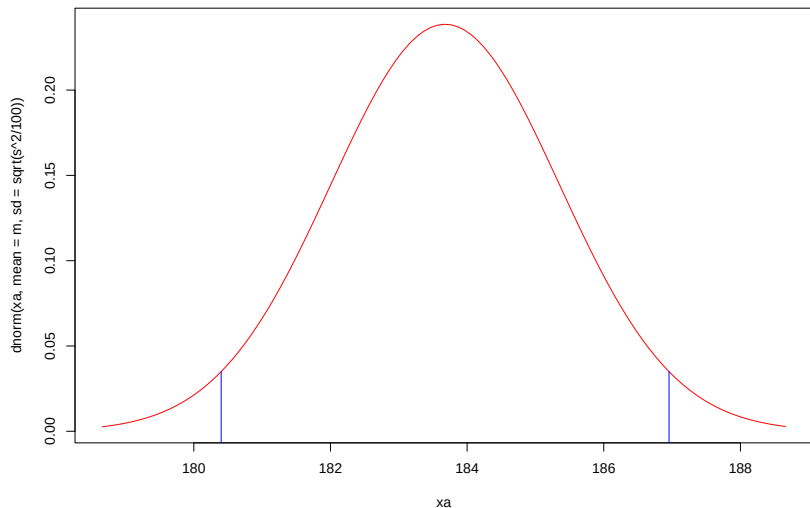
The first sample again

- ▶ $\bar{x} = 183.678$
- ▶ $\hat{\sigma} = 16.721$
- ▶ Our approximation for the sample distribution of $\bar{X}$ is $N(183.678, \frac{16.721^2}{100})$

Estimate of the sampling distribution



95% confidence interval for μ



An interval which covers the true value with probability 0.95.

When can we use this?

- ▶ We have used a large sample approximation, i.e. it is based on the assumption that we have “large enough” sample. What if n is small?
- ▶ In other situations we might be unable to derive the asymptotic distribution, e.g. in situations where the standard central limit theory does not apply.
- ▶ Resampling methods are methods which replicate the idea of sampling from a population described above.
- ▶ The difference is that we sample from our sample instead of from the population. Hence the term resampling.
- ▶ A situation we will often encounter is that we would like to compute $E((Y_i - \hat{f}(X_i))^2)$ and often a resampling method is a good, and feasible, approach to do this.

Cross-validation

- ▶ Cross-validation is the idea of
 - ▶ splitting the data in training and test data
 - ▶ train (estimate) the model on the training data and
 - ▶ evaluate predictions on the test data
- ▶ We will talk about three (or perhaps only two) different approaches
 - ▶ Validation set approach
 - ▶ Leave-one-out cross-validation
 - ▶ k-fold cross-validation

Validation set approach

- ▶ We exemplify with the computation of testMSE
- ▶ Data: $(\mathbf{X}_1, y_1), \dots, (\mathbf{X}_n, y_n)$
- 1. Re-order all observations randomly: $(\mathbf{X}_{(1)}, y_{(1)}), \dots, (\mathbf{X}_{(n)}, y_{(n)})$
- 2. Let the first, say $m = \lfloor n/2 \rfloor$, be the training data

$$(\mathbf{X}_{(1)}, y_{(1)}), \dots, (\mathbf{X}_{(m)}, y_{(m)})$$

and the rest test data

$$(\mathbf{X}_{(m+1)}, y_{(m+1)}), \dots, (\mathbf{X}_{(n)}, y_{(n)})$$

- 3. Fit the model, f to the training data and obtain $\hat{f}$
- 4. Evaluate $\hat{f}$ on the test data

$$\text{testMSE} = \frac{1}{n - m} \sum_{i=m+1}^n (Y_i - \hat{f}(X_i))^2$$

Example on Auto data

```
require(ISLR)
```

```
## Loading required package: ISLR
```

```
names(Auto)
```

```
## [1] "mpg"          "cylinders"    "displacement" "horsepower"   "wei
```

```
## [6] "acceleration" "year"         "origin"       "name"
```

```
n=nrow(Auto)
```

```
draw=sample(1:n,size=floor(n/2))
```

```
head(draw)
```

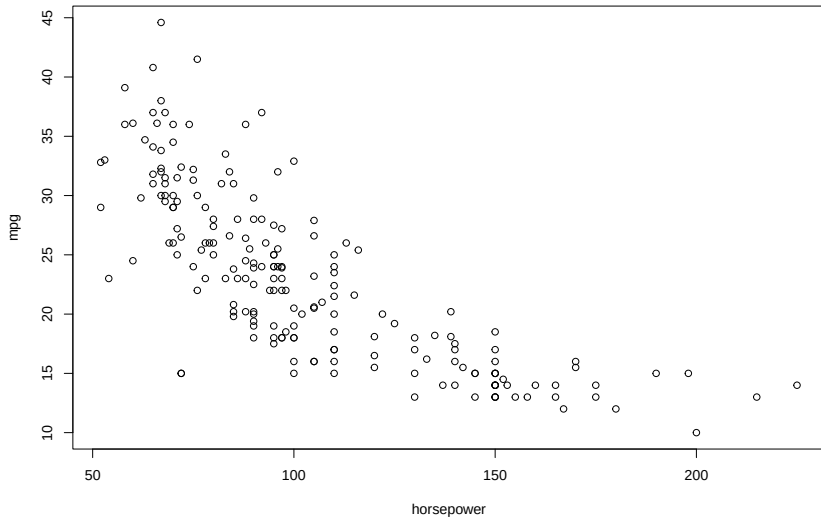
```
## [1] 174 129 361 338 93 189
```

```
train=Auto[draw,] # TRAINING DATA
```

```
test=Auto[-draw,] # TEST DATA
```


mpg against horsepower

```
plot(mpg~horsepower,data=train)
```

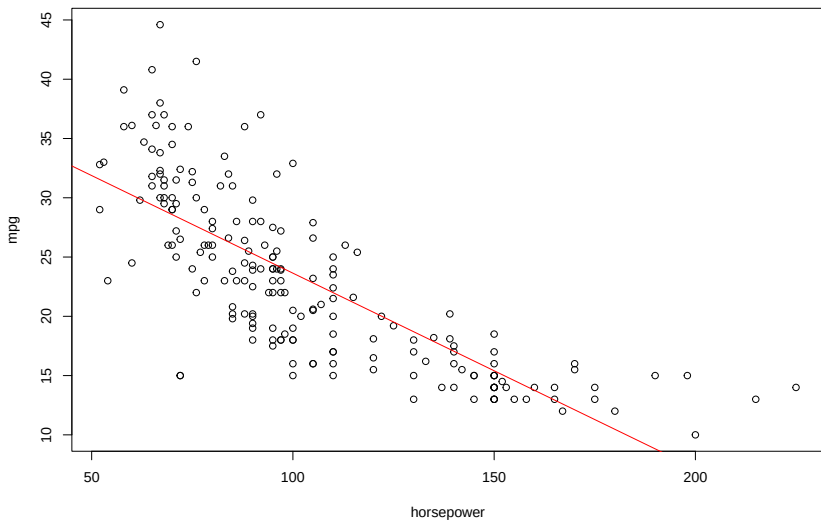


A linear regression

```
mod1=lm(mpg~horsepower,data=train)
summary(mod1)
```

```
##
## Call:
## lm(formula = mpg ~ horsepower, data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.2507  -2.6191  -0.4068   2.7107  15.5260
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  40.106635   1.042605   38.47  <2e-16 ***
## horsepower   -0.164666   0.009551  -17.24  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.634 on 194 degrees of freedom
## Multiple R-squared:  0.6051, Adjusted R-squared:  0.603
## F-statistic: 297.2 on 1 and 194 DF,  p-value: < 2.2e-16
```

```
plot(mpg~horsepower,data=train)  
abline(reg=mod1,col="red")
```



A polynomial regression

```
##
## Call:
## lm(formula = mpg ~ horsepower + I(horsepower^2), data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.3075  -2.1225  -0.1415   2.0981  13.8446
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   56.0674496   2.8042823   19.994 < 2e-16 ***
## horsepower    -0.4598600   0.0495274   -9.285 < 2e-16 ***
## I(horsepower^2) 0.0012249   0.0002023    6.056 7.15e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.259 on 193 degrees of freedom
## Multiple R-squared:  0.6681, Adjusted R-squared:  0.6647
## F-statistic: 194.3 on 2 and 193 DF,  p-value: < 2.2e-16
```

Predicting on test data

```
pred1=predict(mod1,newdata=test)
pred2=predict(mod2,newdata=test)
mse1=mean((test$mpg-pred1)^2)
mse1
```

```
## [1] 27.36073
```

```
mse2=mean((test$mpg-pred2)^2)
mse2
```

```
## [1] 20.29991
```

Did you get the exact same numbers?

Leave-one-out (LOO) CV

-Data: $(\mathbf{X}_1, y_1) \dots, (\mathbf{X}_n, y_n)$

1. Leave the first observation, $i = 1$, out of the sample to form training data $(\mathbf{X}_2, y_2) \dots, (\mathbf{X}_n, y_n)$
2. Fit the model, f to the training data and obtain $\hat{f}$
3. Evaluate $\hat{f}$ on the test data $(\mathbf{X}_1, y_1)$

$$\text{testMSE}_1 = (Y_1 - \hat{f}(X_1))^2$$

4. Repeat 1-3 to obtain testMSE_i for $i = 1, 2, \dots, n$
5. Compute

$$\text{testMSE} = \frac{1}{n} \sum_{i=1}^n \text{testMSE}_i$$

Auto data

```
# LEAVE-ONE-OUT
MSE1=matrix(0,n,1)
MSE2=matrix(0,n,1)
for(i in 1:n)
{
  train=Auto[-i,]
  test=Auto[i,]
  mod1=lm(mpg~horsepower,data=train)
  mod2=lm(mpg~horsepower+I(horsepower^2),data=train)
  pred1=predict(mod1,newdata = test)
  pred2=predict(mod2,newdata = test)
  MSE1[i]=(test$mpg-pred1)^2
  MSE2[i]=(test$mpg-pred2)^2
}
```

MSE1 = 24.23

MSE2 = 19.25

k-fold CV

- ▶ k groups of size $\frac{n}{k}$
- ▶ “Leave- $\frac{n}{k}$ -out CV”
- ▶ $k = n =$ Leave-one-out
- ▶ Choose smaller k if the method is computationally demanding

Pros and cons with the different approaches

	Validation set	LOO	k-fold
Estimation uncertainty	Worst	Best	Intermediate
Computational time	Best	Worst	Intermediate

There are also bias-variance considerations to be made. More on that on the next tutorial.